

Linking root traits and soil moisture redistribution under *Achnatherum splendens* using electrical resistivity tomography and dye experiments

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ABSTRACT

The rooting traits of deep-rooted alpine grasses strongly control the redistribution of soil moisture (i.e., the changes in the soil volumetric water content, Δ SVWC), which is the most important water resource supporting vegetation growth in cold semiarid regions. However, due to the difficulty of developing appropriate methods to characterize the hidden structures of root-soil systems, the linkages between root traits and Δ SVWC remain challenging to detect. This study aims to identify how the roots of *Achnatherum splendens* (*A. splendens*) affect soil water infiltration and spatiotemporal variations in the rhizosphere under different root densities and rainfall patterns using electrical resistivity tomography (ERT) and dye experiments, as well as to determine the relationship between root traits and soil water redistribution. Manipulative rainfall experiments 5, 10, 20 and 40 mm simulated rainfall were applied to sample plots with no plants (control), single and double *A. splendens* plantings (BAS, SAS, DAS, respectively). The results of ERT and the dyeing profiles showed that the roots of *A. splendens* greatly affected infiltration processes and the soil water distribution. The soil water was vertically and laterally funneled along the roots. In SAS, which had a lower root length density (RLD), the soil water accumulated mainly in the shallow layer (0–20 cm), while in DAS, which had a higher total root mass density (RMD), the soil water mainly infiltrated rapidly into the deep soil (>20 cm), especially during the extreme (40 mm) rainfall event. The temporal increases in soil moisture below 20 cm under DAS were higher than that those under BAS and SAS after rainfall for 2 and 24 h. The RMD, specific root length (SRL), specific root surface area (SSA) and RLD were all negatively and significantly correlated with the Δ SVWC at 0–20 cm. At 20–40 cm, the RMD and RLD were positively and significantly correlated with Δ SVWC, whereas the SRL and SSA showed the opposite trend. These findings provide direct evidence of the linkages between *A. splendens* root traits and Δ SVWC using ERT and the dye tracing method, and this evidence is important for the development of soil-plant-atmosphere continuum (SPAC) models.

1. Introduction

Water is the key factor controlling vegetation growth in grassland ecosystems, and ecological-hydrological processes in subsurface environments are particularly important (Li, 2011; Tilman, 1998). Precipitation provides most of the soil water that sustains plant growth in arid

and semiarid regions (Wang et al., 2008; Dahan et al., 2007). Root traits have decisive effects on the life of plants and their adaptivity to the environment (Bardgett et al., 2014). Therefore, exploring the linkage between root traits and soil moisture redistribution (i.e., changes in the soil volumetric water content, Δ SVWC) is important for understanding interactions in the soil-plant-atmosphere continuum (SPAC, Tramontini

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et al., 2013).

The spatiotemporal dynamics of soil moisture in the rhizosphere contribute to understanding the hydrological processes associated with vegetation, as well as the appropriate design and calibration of SPAC models (Vereecken et al., 2013; Western et al., 2003). Soil water is funneled along roots, forming preferential flow that triggers rapid lateral subsurface flow and creates nonuniform flow transport, as has been demonstrated in previous studies (Schwartzel et al., 2012; Levia and Frost, 2003). Many studies have focused on differences in soil moisture in the rhizosphere among different species of plants and/or the corresponding physiological characteristics of root systems (Koch et al., 2018; Ain-Lhout et al., 2016; Srayeddin and Doussan, 2009). In addition, soil physical properties always serve as the main indicators of the indirect relationships between roots and Δ SVWC. High-density root systems and root channels improve soil infiltrability and hydraulic conductivity by altering soil physical properties, such as macropores, bulk density and soil particles size (Guo et al., 2019; Hu et al., 2019; Wu et al., 2017; Van Schaik, 2009). The correlation between soil infiltration and the roots at the point and column scale was also studied. Specific studies have shown that the infiltration of soil moisture is positively related to a variety of root structural traits, such as root dry weight density (Huang et al., 2019), root biomass and length (Liu et al., 2019), belowground biomass (Huang et al., 2017), root surface area density (Li et al., 2013) and root network density (Hu et al., 2019). However, further investigation is needed to provide evidence of direct linkages and dynamic interactions between root traits and soil moisture dynamics. The heterogeneity of soils and the difficulty of observing soil structures hamper the study of spatiotemporal changes in Δ SVWC (Allroggen et al., 2015). Due to the limitations of the relevant methods and technology, verifying whether roots can affect infiltration in the soil or the spatiotemporal variations in rhizosphere soil moisture will require further exploration.

Soil moisture observation is the key to characterizing earth surface processes. Traditionally, researchers have used gravimetric analysis of extracted samples or sensors to measure soil moisture. The accuracy of gravimetric analysis is limited by the lack of sampling points and its disturbance of the observed soil (Sperling and Lazarovitch, 2010). It is difficult to observe soil hydrological processes across spatiotemporal scales using the above methods (Doolittle and Brevik, 2014; Tetzlaff et al., 2014; Vereecken et al., 2008). In addition, preferential flow along heart and sinker roots (Li et al., 2009) and laterally oriented coarse roots (Liang et al., 2011) as well as through decayed root channels in the soil (Spencer and van Meerveld, 2016; Schwartzel et al., 2012) have been visualized by dye staining experiments in previous studies. However, this method is destructive and consequently nonrepeatable (van Schaik, 2009), which makes it difficult to quantify spatiotemporal variations in soil moisture. Electrical resistivity tomography (ERT), which is a minimally invasive and scalable imaging geophysical method of providing data with high spatial and temporal resolution (Al Hagrey, 2007; Michot et al., 2003), is considered one of the most effective geophysical methods (Vanella et al., 2018) of imaging the subsurface and acquiring its associated properties. To date, ERT has been used successfully in measuring the temporal dynamics and spatial distribution of the subsurface soil moisture and the routing of stemflow (Guo et al., 2020; Dick et al., 2018; Ain-Lhout et al., 2016; Fan et al., 2015). Nevertheless, ERT is limited by uncertainty in data interpretation because the measurements taken in the soil and root zones are indirect (Cimpoiasu et al., 2020). Therefore, ERT should be integrated with traditional methods such as dye tracing to enhance the reliable characterization of subsurface ecohydrological processes.

The Qinghai Lake Basin (QLB) is located in the cold semiarid zone on the northeastern Qinghai-Tibet Plateau (QTP) which is sensitive to climate and global environmental change (Li et al., 2018). Recently, the entire watershed has faced serious ecological damage and environmental degradation due to climate change and human activities (Li et al., 2018). The deep-rooted grass *Achnatherum splendens* (*A. splendens*) is the representative species of the warm steppe community in the QLB

(Huai et al., 2008). *A. splendens* has strong adaptability to diverse environments and a strong tolerance to salt, alkali, drought and cold; this strong tolerance plays a critical role in ecological security in the QLB (Miehe et al., 2008; Huai et al., 2008). On the QTP, climate change has increased precipitation variability, with more frequent droughts and extreme humid conditions (Yang et al., 2011). Specifically, the *A. splendens* steppe has degenerated rapidly due to climate change and human activity (Li et al., 2007). The annual evapotranspiration on the QTP exceeds the total precipitation, especially in the growing season (Zhang et al., 2016). Therefore, water is a key factor limiting vegetation growth in the *A. splendens* steppe ecosystem. It has been shown that *A. splendens* patches can affect the spatial distribution of soil moisture and salinity, and absorb moisture to form a “water island” (Jiang et al., 2017). The root system plays an important role in water accumulation under *A. splendens* patches. Previous studies have revealed that individual *A. splendens* plants have well-developed roots in the 0–10 cm soil layer (Chen et al., 2017). A large amount of macropores was also found in the 0–20 cm soil layer under *A. splendens* patches, which can be attributed to fast root development in area where rain infiltrates more easily into the soils (Jiang et al., 2017; Hu et al., 2016). The Δ SVWC under *A. splendens* patches is also affected by rainfall patterns. Jiang (2016) observed that the response time and depth of soil moisture were shorter and deeper, respectively, with increasing rainfall amount under *A. splendens* patches. However, few studies have been conducted on the dynamic changes in soil moisture under *A. splendens* at different root densities after different amounts of rainfall, and the impact of the roots on moisture transport is still unknown.

To bridge these knowledge gaps, this study aims to identify how roots of *A. splendens* affect water infiltration and spatiotemporal soil moisture variations in the rhizosphere under different root densities and rainfall patterns using ERT and dye experiments, as well as to determine the relationship between root traits and Δ SVWC. Therefore, the specific objectives of this study are (1) to identify whether *A. splendens* roots can affect infiltration in the soil; (2) to describe the spatiotemporal characteristics of the Δ SVWC of *A. splendens* under different root densities and rainfall patterns; and (3) to reveal the relationships between *A. splendens* root traits and Δ SVWC.

2. Material and methods

2.1. Experimental sites

The study was conducted in the QLB, which is situated in the northeastern QTP, and is a semiarid, cold, high-altitude climatic area (Fig. S1a) (Li et al., 2018). *A. splendens* is the dominant species in the temperate steppe ecosystem in the QLB, and it is mainly distributed along Qinghai Lake boundaries (Li et al., 2018). We selected sampling areas from areas of *A. splendens* steppe in the northern QLB (37°14′51.02″N, 100°14′15.31″E; Fig. S1b). The topography is relatively flat, with an elevation of approximately 3205 m (Jiang et al., 2017). The local climate is a semiarid temperate continental climate, with a mean annual temperature of -3.3°C (Jiang et al., 2017). The mean annual precipitation is 378.2 mm, with approximately 70%–80% occurring from July to October (Jiang et al., 2017). The zonal soils are mainly chestnut soil developed from Mol-Cryic Cambisols, according to the Chinese Classification (Cooperative Research Group on Chinese Soil Taxonomy, 2001), or Calcicorthic Aridisols, according to the U.S. Soil Taxonomy (Soil Survey Staff, 1987). Four reference soil layers were determined as follows (Fig. S1c): (1) the humus layer (A, 0–5 cm); (2) the eluvial layer (E, 5–40 cm); (3) the illuvial layer (B, 40–105 cm); and (4) the parent material (C, >105 cm). The illuvial layer is subdivided into two parts: B₁ (5–40 cm) and B₂ (40–80 cm), based on an increase in sand content. The basic soil physical properties of the sample plots determined through a field survey are shown in Table 1 (e.g., soil texture, bulk density, soil particle composition, soil total porosity, saturated hydraulic conductivity). There is a distinct soil horizon

Table 1

The basic physical properties of the soil in the sample plots.

Soil layer	Soil depth (cm)	Sand	Silt	Clay	Bulk density (g/cm ³)	Soil total porosity (%)	Saturated hydraulic conductivity(mm/min)
A	0–5	54.11%	39.03%	6.87%	1.15	50.22	2.21
E	5–40	51.15%	36.63%	12.23%	1.17	48.82	1.22
B ₁	40–80	53.37%	41.47%	5.17%	1.18	49.77	0.90
B ₂	80–105	72.04%	22.67%	5.29%	1.70	25.84	1.37
C	>105	65.53%	28.48%	5.99%	1.75	7.96	1.27

boundary at 40 cm depth because the saturated hydraulic conductivity is considerably lower below this depth (Table 1). The soil particle composition was determined using a Mastersizer 3000 (Malvern Inc, UK). The saturated hydraulic conductivity was measured by the cylindrical soil cutter method. The main soil parent material types are sandstone and siltstone.

2.2. Design and measurements of the manipulative rainfall and dye tracing experiments

The rainfall events were classified by the amount of rainfall (Schwinning and Sala, 2004), and the rainfall gradient was divided into 5 levels (0–5 mm/5–10 mm/10–15 mm/15–20 mm/>20 mm). According to Jiang's findings (2016), from 2013 to 2015, the average rainfall frequency and average rainfall in the study area were 47 times those of small rainfall events (0–5 mm), and the annual rainfall amount from small rainfall event was 71.9 mm. There were 19 moderate rainfall events (5–15 mm), resulting in 162.4 mm of rainfall. Heavy rainfall events (15–20 mm) occurred three times, resulting in 59.4 mm of rainfall. Extreme rainfall events (>20 mm) occurred three times, resulting in 67.3 mm of rainfall. Therefore, four levels for the manipulative rainfall events (5, 10, 20 and 40 mm) were used in this study. During August 2019, 24 rainfall simulation experiments (4 rainfall levels $\times$ 3 root densities $\times$ 2 replications) were conducted. The different root density plots contained no plants (blank *A. splendens* (BAS)), single *A. splendens* plants (SAS) and double *A. splendens* plants (DAS). The BAS and SAS sample plots were delimited to 1 $\times$ 1 m by a colored steel frame, and the DAS sample plot was 2 $\times$ 1 m. The steel frames were painted on both sides to have less of an effect on the ERT measurement. The rainfall simulator was a needle-type device with a 1 $\times$ 1 m square gauge. There were 761 needles distributed on the bottom of the gauge (staggered arrays with 20 $\times$ 20 odd rows and 19 $\times$ 19 even rows). The distance between needles was 5 cm in both the horizontal and vertical directions and 4 cm in the diagonal direction. Other detail about the device can be found in Wang et al. (2019). The BAS and SAS treatments each used a rainfall simulator that started simultaneously raining along a straight line (Fig. S2a); the DAS treatment used two rainfall simulators together (Fig. S2b). All sample plots were subjected to a two-day controlled drought period prior to the simulated rainfall and to rainproofing measures immediately after the manipulative rainfall experiment ended. The aboveground parts of the *A. splendens* plants were cut to ground level to avoid any effect on rainwater infiltration when the manipulated rainfall occurred.

The average discharge of the simulated rainfall device used in this experiment is approximately 0.33 mm/min, which is much lower than the saturated hydraulic conductivity of the 0–5 cm soil layer (Table 1). The dye tracing experiments in the plots that had no surface water were carried out at the same time as the manipulative rainfall experiments. The concentration of brilliant blue dye in the artificial rainwater was 5 g/L. The infiltration depth reached the maximum soil depth for the set rainfall amount after 24 h of simulated rainfall. The pre-experimental observation showed that the infiltration depth after 68 h rainfall was lightly different from that after 24 h. Hence, after 24 h, a soil profile of 0.4 m was dug out from the edge of the steel frame. Photographs were taken with the photographic line of sight perpendicular to the soil profile. The dyeing profile data were processed as follows. First, the images

of vertical profiles taken in the field were processed with Photoshop CC 2018 software. The dyed section was replaced with black, while the unstained soil appeared white. Then, the bitmaps of the images were processed by MATLAB2016b (MathWorks Inc, USA), and binary images were output. The maximum infiltration depth (MID, m) is the maximum depth of the brilliant blue dye, representing the maximum water penetration depth. The area per unit depth (AD, %) is the ratio of the area stained by brilliant blue dye to the total area obtained per centimeter depth. All the above indicators were calculated with MATLAB2016b software. For detailed information, refer to Weiler and Fluhler (2004).

2.3. ERT measurements, data inversion and field calibration

2.3.1. ERT data collection and measurements

The ERT data were collected by a Terrameter LS 2 electrode system (ABEM instruments, Sundbyberg, Sweden). The Wenner array was used for all surveys because of its high vertical resolution and signal-to-noise ratio (Zhu et al., 2007; Samouëlian et al., 2005). Previous studies have found that the root system of *A. splendens* at the experimental site is mainly distributed at a depth of 0–80 cm (Jiang, 2016). When the length of the ERT survey line was 6 m, the detected soil depths were 0.075, 0.225, 0.383, 0.556, 0.746 and 1.187 m, which met the requirement of this study. The maximum number of stackings used for measurement was 4. The error limit of measurement, also called the variation coefficient of repeated measurements/stackings, was 1%. For the plot surveys, we established 6 survey lines separated by 0.2 m. The plot size was 6 $\times$ 1 m (the same as the survey grid), and the spacing between electrodes was 0.3 m, which is the acceptable minimum spacing for this instrument. Twenty stainless steel electrodes (0.8 cm in diameter, 40 cm length, extending 5 cm into the soil) were inserted into the soil along each survey line for the ERT measurements. A total of 80 ERT experiments (4 rainfall levels $\times$ 2 sample ERT plots $\times$ 2 replications $\times$ 5 time points) were designed for this study; the BAS and SAS treatments were measured on the same line. Reference ERT datasets were obtained prior to the manipulative rainfall experiments. The electrical resistivity (ER) of several vertical soil sections was measured out after the manipulative rainfall experiment at intervals of 2, 6, 12 and 24 h. However, when natural rainfall occurred after the simulated rainfall, the ERT measurements were interrupted, leading to a substantial amount of data loss. In total, 62 ERT datasets were collected.

2.3.2. Electrical resistivity inversion

Panissod et al. (2008) showed that two-dimensional resistivity inversion is more appropriate than three-dimensional inversion for reflecting the soil moisture content in profiles. RES2DINVx64 software (ver. 4.05) is a computer program that can automatically generate a two-dimensional resistivity model from electrical imaging surveys. The software parameters used in this study are shown in Table S2. The inversion routine applied in this study is based on the smoothness-constrained least-squares method (De Groot-Hedlin and Constable, 1990). This minimizes the square of the changes (L2 norm) in the modelled resistivity values to produce a model that exhibits smooth variation in resistivity values. The equation is as follows:

$$(J^T J + \lambda F) \Delta q_k = J^T g - \lambda F q_k \quad (1)$$

where F is a smoothing matrix, $F = \alpha_x C_X^T C_X + \alpha_z C_Z^T C_Z$; C_X is the hori-

zontal roughness filter; C_z is the vertical roughness filter; J is the Jacobian matrix of partial derivatives; J^T is the transpose of J ; λ is the damping factor; Δq_k is the change in model resistivity; and g represents the data misfit, showing the difference between the calculated and measured apparent resistivity values. The value in g is given by the root-mean-squared error (RMSE), which is the quantitative indicator of the fit of each iteration that is used to find the best model. The lowest possible RMSE would indicate that there are large and unrealistic variations in the ER values. Thus, we set the number of iterations to 7. The L-curve method was used to determine the optimum λ . This method essentially assumes that the data misfit and the model roughness change in a smooth manner with the damping factor to obtain a unique point of maximum curvature (Farquharson and Oldenburg, 2004). Finite-difference forward modeling was used to calculate the apparent resistivity values in this study; this method is faster than others because the dataset in this study does not contain topographical data.

There are two ways to solve the least-squares equation in RES2DINVx64 software. We used the incomplete Gauss–Newton method to reduce the inversion time. This method can improve the convergence of the iterative method by scaling the matrices. Additionally, a constraint was added to Eq. (1) in order to incorporate a reference model (Farquharson and Oldenburg, 2004). The equation is as follows:

$$(J^T J + \lambda(F + \mu I)) \Delta q_k = J^T g - \lambda F q_k - \lambda \mu (q_k - q_m) \quad (2)$$

where q_m is a reference model, usually a homogeneous half-space model; and μ is an additional damping factor to control the degree when the inversion model is close to q_m . The value of μ is normally between 0.01 and 0.10; we set μ to 0.01.

Before inversion, bad data points were first removed manually. Noise estimates for the reciprocal measurements in the field surveys were not performed due to time constraints. However, a diagonal filter was used to modify the roughness filter so that it also had components in the diagonal direction (Farquharson, 2008). In addition, we calculated the resolution and sensitivity of ER in order to control the data quality. The average of all the relative misfit values for all the inversions was 2.99%, with minimum and maximum values of 0.48% and 20.49%, respectively. The coefficient of variation was 0.99.

2.3.3. ERT sensitivity and resolution

The cumulative sensitivity, which evaluates the effect of the model parameters on the modeled measurements, was used in this study (Henderson et al., 2009). The integrated sensitivity value associated with each model block was provided by RES2DINVx64. The equation is as follows:

$$S_i = \frac{1}{m} \sum_{j=1}^m |J_{ij}| \quad (3)$$

where S_i is the sum of the Jacobian matrix values associated with the model block for all the data points. The higher the sensitivity value is, the more reliable the model resistivity value is. The surface blocks usually have higher sensitivity values than other blocks because the values of the sensitivity function are higher near the electrodes.

The model resolution is also an indicator of data quality (David Caterina et al., 2013). The model resolution matrix is calculated as follows:

$$R = (J^T J + \lambda F)^{-1} J^T J \quad (4)$$

The model resolution of the blocks is expressed by the diagonal elements in the matrix. R is the model resolution matrix, which is used as a filter to resolve the subsurface resistivity (Day-Lewis et al., 2005). The damping factor in the inversion least-squares method is normally increased for each deeper soil layer to obtain a stable inversion process, because the resolution of the resistivity method decreases exponentially with depth. The average relative sensitivity of all the measurements was

1, with minimum and maximum values of 0.0056 and 10.3613, respectively. The average smoothed sensitivity of all the measurements was 1.0048, with minimum and maximum values of 0.0042 and 9.3934, respectively. The average model resolution of all the measurements was 0.1571, with minimum and maximum values of 0.0105 and 0.8137, respectively.

2.3.4. Temperature correction for resistivity

Temperature is an important factor affecting soil resistivity. A soil profile was dug to identify the characteristics of the soil and install soil temperature probes. Nine soil temperature probes (Campbell Sci. 109) were installed at different depths in the soil (0.05, 0.1, 0.2, 0.4, 0.8, 1.2, 2, 3 and 5 m). The straight distance range between the locations of the various sample plots and the soil temperature probes were 49–102 m (Fig. S3). After the sensors were installed, the soil profile was carefully refilled layer by layer. To eliminate the effect of the temperature and determine only the relationship between the ER and the soil moisture in each layer, all the ER measurements were calibrated and scaled to a reference temperature of 25°C (Keller and Frischknecht, 1966) as follows:

$$\rho_{25} = \rho_T (1 + \alpha \times (T - 25)) \quad (5)$$

where ρ_T denotes the electrical resistivity at temperature T ; ρ_{25} is the electrical resistivity at $T = 25^\circ\text{C}$; and α is an empirical coefficient that often equals 0.025°C^{-1} .

2.3.5. Field calibration and evaluation

Michot et al. (2003) confirmed that considering the soil layer could allow the petrophysical relationship between the change rate of soil moisture and ER to be established more accurately. Therefore, we performed an “in situ” calibration between the measured soil moisture and the ER of typical soil layers. A total of 960 soil samples were collected to construct the petrophysical relationship. Before rainfall, 384 soil samples were obtained (4 rainfall levels $\times$ 2 treatment replications $\times$ 3 sampling replications $\times$ 8 soil depths $\times$ 2 ERT sample plots) along the ERT survey lines. After rainfall for 24 h, 576 soil samples (4 rainfall levels $\times$ 3 root densities $\times$ 2 treatment replications $\times$ 3 sampling replications $\times$ 8 soil depths) were collected along the ERT survey transects. The soil samples were collected at depths of 5, 10, 15, 20, 30, 40, 60 and 80 cm. The positions of the soil samples in the horizontal direction are shown in Table S1. The bulk density was determined by the cylindrical soil cutter method. Three ring knife samples were collected in each soil layer from the excavated soil profile, and total of 21 ring knife soil samples were collected (3 sampling repeats $\times$ 7 soil layers in all, Fig. S1c). The sample plots are close to the location of the soil temperature probes (Fig. S3), so we assumed a constant bulk density for a given horizon. The soil volumetric water content (SVWC) (cm^3/cm^3) was calculated as the product of the soil bulk density (g/cm^3) and the gravimetric soil water content (SWC, g/g) and was used to construct the petrophysical relationship. Considering the effect of brilliant blue dye on ER in sample plots after rainfall (Koestel et al., 2009), we constructed the relationships between SVWC and ER before and after rainfall. The petrophysical relationships were developed on the basis of the SVWC of the soil samples and the ER value determined through ERT at the corresponding position. Regression analysis was performed in the program R (x64 3.5.1) to determine the petrophysical relationship. Residual analysis and Cook’s distance (an indicator of leverage) were used to remove outlier SVWC values before the regression analysis. After the data quality control process, a total of 745 data pairs were used to explore the relationships between SVWC and ER. Eighty percent of the dataset was randomly selected to build models, and the remaining 20% was used to verify the accuracy of the models. The Pearson correlation coefficient (r) and RMSE between the predicted and measured SVWC values were used to evaluate the accuracy of the model:

$$r = \frac{\sum_{i=0}^n (MSVWC_i - \overline{MSVWC}) (PSVWC_i - \overline{PSVWC})}{\sqrt{\sum_{i=0}^n (MSVWC_i - \overline{MSVWC})^2 \sum_{i=0}^n (PSVWC_i - \overline{PSVWC})^2}} \quad (6)$$

$$RMSE = \sqrt{E((PSVWC - MSVWC)^2)} \quad (7)$$

where r is the correlation coefficient; n is the number of test datasets; $MSVWC_i$ is the i th measured SVWC in these datasets; $PSVWC_i$ is the i th predicted SVWC in these datasets; and $\overline{MSVWC}$ and $\overline{PSVWC}$ are the sample means of the measured and predicted SVWC, respectively, for these datasets. $E()$ is the formula used to calculate the mean value. A higher correlation coefficient and a lower RMSE indicate higher model accuracy.

2.4. Measurement of root sampling

After 24 h of rainfall, a root auger with an internal diameter of 8.2 cm was used to collect soil cores. Three soil cores were collected from each of the sample plots of different root densities and each soil core was sampled in 10 cm increments to 80 cm depth, i.e., at depths of 10, 20, 30, 40, 50, 60, 70 and 80 cm. The location where the root samples were taken are shown in Figs. S4 and S5. A total of 384 root samples (4 rainfall levels $\times$ 2 ERT sample plots $\times$ 2 treatment replications $\times$ 3 sampling replications $\times$ 8 soil depths) were collected in this study. The root were

treated as follows: (1) the collected roots were washed with clean water to remove dust and sediment; (2) roots from each depth were randomly selected for scanning using a scanner (EPSON Perfection V37/V370); (3) the root length and surface area of each layer were measured with WinRHIZO software (Regent Instruments, Canada Inc.); (4) the scanned root samples were placed into envelopes and heated in an oven at 105 °C for 15 min and then at 70 °C until they dried to constant weight (Zhang et al., 2009); and (5) the root biomass was obtained by weighing. The root traits were then calculated according to the root length, root surface area and soil core volume (Erktan et al., 2016). The total root mass density (RMD), root length density (RLD), specific root length (SRL) and specific root surface area (SSA) are important root traits used to describe root morphology (Fitter, 1985). RMD is the root biomass per unit volume and is calculated as the ratio between the total root dry mass and the soil volume from which the roots were extracted. The SRL is the root length per unit of root biomass and is calculated as the ratio between the total root length and the root dry mass. The SSA is the surface area per unit of root biomass and is calculated as the ratio between the total root surface and the root dry mass. These traits indicate the ability of roots to acquire soil resources (de Vries et al., 2016; Poorter and Ryser, 2015). The RLD is the total root length per unit of soil volume which can be used to assess root distribution traits in soil such as the root extension and the interlacing or interweaving of roots in soil (Kato et al., 2006); the RLD is calculated as follows: $RLD = RMD \times SRL$. We selected the data set obtained before the manipulative rainfall experiments as the

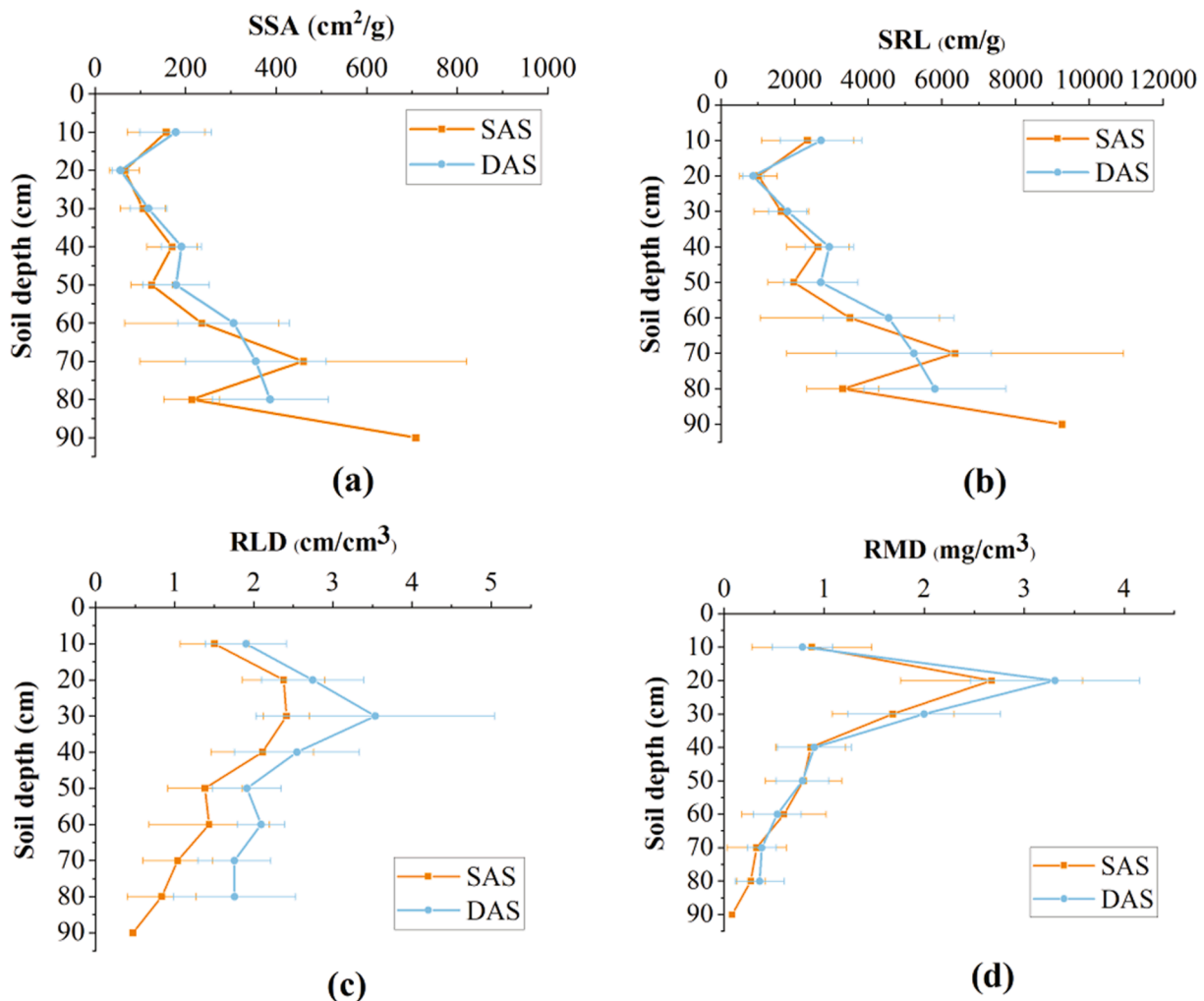


Fig. 1. Vertical distributions of *A. splendens* root traits at different root densities. a: specific root surface area (SSA); b: specific root length (SRL); c: root length density (RLD); d: total root mass density (RMD). BAS is blank *A. splendens*, SAS is single *A. splendens* and DAS is double *A. splendens*.

reference time. The change rate of the soil volumetric water content (Δ SVWC) was calculated as follows: Δ SVWC = (the SVWC after rainfall of 2,6,12 or 24 h - the SVWC before rainfall)/the SVWC before rainfall. Here, the SVWC before and after rainfall was estimated based on the ERT data and the petrophysical relationships. The Δ SVWC was calculated for only the root zone (Figs. S4 and S5).

3. Results

3.1. Root traits

The vertical distributions of the root traits of *A. splendens* in plots different root densities are shown in Fig. 1. The SRL and SSA showed a consistent trend with increasing soil depth. The SRL and SSA were relatively high at the surface layer (10 cm), reached their minimum values (874.92 cm/g and 55.97 cm²/g, respectively) at 20 cm and then increased with increasing depth below 20 cm (Fig. 1a, b). The maximum values of RLD and RMD were 3.53 cm/cm³ at 30 cm and 3.31 g/cm³ at 20 cm, respectively. The RLD and RMD increased with increasing soil depth at 0–30 cm and 0–20 cm, respectively, whereas they showed the opposite pattern below 30 cm and 20 cm (Fig. 1c, d). The SRL and SSA of DAS were larger than those of SAS at a soil depth of 40–60 cm (Fig. 1a, b). In the whole soil profile, the RLD of DAS was larger than that of SAS, and the largest difference was observed at 30 cm. The RMD of DAS and SAS showed a sudden increase at 20 cm, and the difference between them was the largest at 20 cm.

3.2. Characteristics of infiltration obtained from the dyeing profile

Our results showed that the order of the AD at the same soil depth in the different root-densities treatments was DAS > SAS > BAS at 0–20 cm and DAS > BAS > SAS below 20 cm (Fig. 2a). These results indicate that the water infiltration in DAS was stronger than that in SAS and BAS in both the horizontal and vertical directions. In addition, the water infiltration in SAS was stronger than that in BAS only at 0–20 cm. The MID ranges (namely, the soil depth at which AD was zero) in BAS, SAS and DAS were 0.11 (5–2) ~ 0.34 m (40–1), 0.13 (10–1) ~ 0.25 m (40–2) and 0.12 (5–2) ~ 0.38 m (40–1), respectively (Fig. 2b, c, d). These results

Table 2

The petrophysical relationships between the soil volumetric water content (y) and electrical resistivity (x) of two soil layers before and after simulated rainfall.

Statistical indicators	Before rainfall 0–40 cm	Before rainfall >40 cm	After rainfall 0–40 cm	After rainfall >40 cm
	$y = a \cdot x^2 + b \cdot x + c$	$y = a + b \cdot \log x$	$y = a \cdot (x^b)$	$y = a + b \cdot \log x$
R ²	0.4918	0.4870	0.4302	0.5770
RMSE	0.0275	0.0282	0.0449	0.0250
a	0.000045	0.2707	0.4058	0.3024
b	−0.004274	−0.0330	−0.1106	−0.0367
c	0.2921	–	–	–
P	<0.001	<0.001	<0.001	<0.001

Note: P, level of significant difference

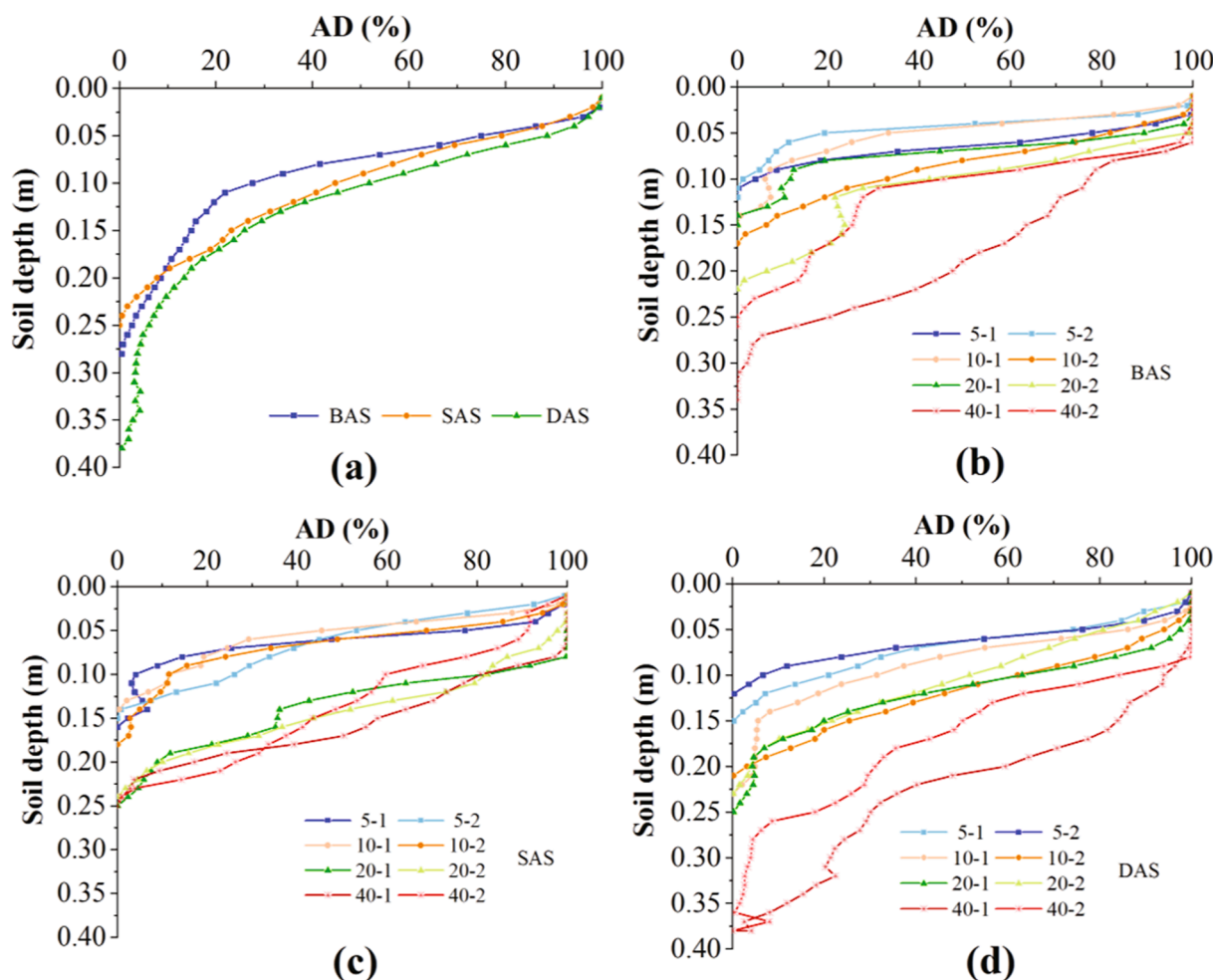


Fig. 2. Variation in dyeing area per unit depth (AD) along the soil profile under (a) average rainfall patterns; (b) blank *A. splendens* (BAS); (c) single *A. splendens* (SAS); (d) double *A. splendens* (DAS).

show that the MID in the BAS, SAS and DAS treatments occurred under the 40 mm rainfall and that the MID of the DAS treatment was the highest.

3.3. Spatiotemporal variations in soil moisture redistribution determined with ERT data

The petrophysical relationships between the ER and Δ SVWC of two typical horizons (0–40 cm and > 40 cm) before and after simulated rainfalls were obtained by comparing various regression models (Table 2 and S3). The soil moisture data estimated by ERT data were effectively correlated with the measured values, and r ranged from 0.5708 to 0.8752 (Fig. 3). Figs. 4, S6, S9 and S12 show the dynamic Δ SVWC of *A. splendens* in vertical transects and dyeing profiles under different root densities and rainfall patterns. Horizontally, soil depths of 7.5 and 22.5 cm were selected as examples to show the Δ SVWC (Figs. 5, 6, S7, S8, S10, S11 and S13). In the vertical and horizontal profiles, the soil moisture was obviously concentrated in the rhizosphere in both SAS and DAS. In addition, areas with greater proportions of dye corresponded to various root positions belowground (Figs. 4j–l, S6i–k, S9j–l and S12j–l). These results suggest that the soil moisture flowed along the root system and that, as a result, most of the wetted areas appeared in the rhizosphere.

The variations of Δ SVWC with time in BAS under different rainfall patterns were uniform and were $21.83\% \pm 0.1990 \sim 26.88\% \pm 0.1682$ and $15.62\% \pm 0.2928 \sim 23.74\% \pm 0.1602$ in the cutting planes of 7.5

and 22.5 cm, respectively (Figs. 5, 6, S7, S8, S10, S11 and S13; Table 3). The increase in SVWC in BAS decreased with increasing soil depth, and the increase in SVWC in SAS and DAS was mainly concentrated in the surface layer (7.5 cm) and deeper layer (38.3 cm) after rainfall for 24 h (Figs. 5, 6, S7, S8, S10, S11 S13; Table 3). The mean increase in Δ SVWC at a depth of 7.5 cm after rainfall for 2 and 24 h was the highest in SAS, at $22.30\% \pm 0.1584$ and $23.30\% \pm 0.1576$, respectively (Table 3). In addition, the mean increases in Δ SVWC at 22.5 and 38.3 cm in DAS were $21.72\% \pm 0.0918$ and $26.07\% \pm 0.0244$, respectively, which were higher than those in BAS and SAS after 24 h of rainfall (Table 3). Similar results were also obtained in DAS after rainfall for 2 h (38.3 cm, $24.83\% \pm 0.0316$) (Table 3). Based on the above results, more soil moisture accumulated in the shallow layer (<20 cm) under SAS than under BAS and DAS. Under DAS, soil water infiltrated rapidly (within 2 h) to the deeper soil (>20 cm), and the duration of infiltration lasted for 24 h.

The mean Δ SVWC at 38.3 cm in DAS, $28.26\% \pm 0.0145$, was higher than that in BAS and SAS, whereas the increase rate at 7.5 cm in SAS, $37.51\% \pm 0.0419$, was higher than that in BAS and DAS under 40 mm simulated rainfall (Table 3). This result may indicate that 40 mm of rainfall further facilitated accumulation of water in the shallow soil in SAS and the infiltration of water into the deep soil in DAS. In this paper, only one set of results is shown in the figures due to considerations of paper length. Fig. 7 shows the average relative error between two duplicated experiments; the repeated results show high reproducibility, with a total average relative error of 3.61%.

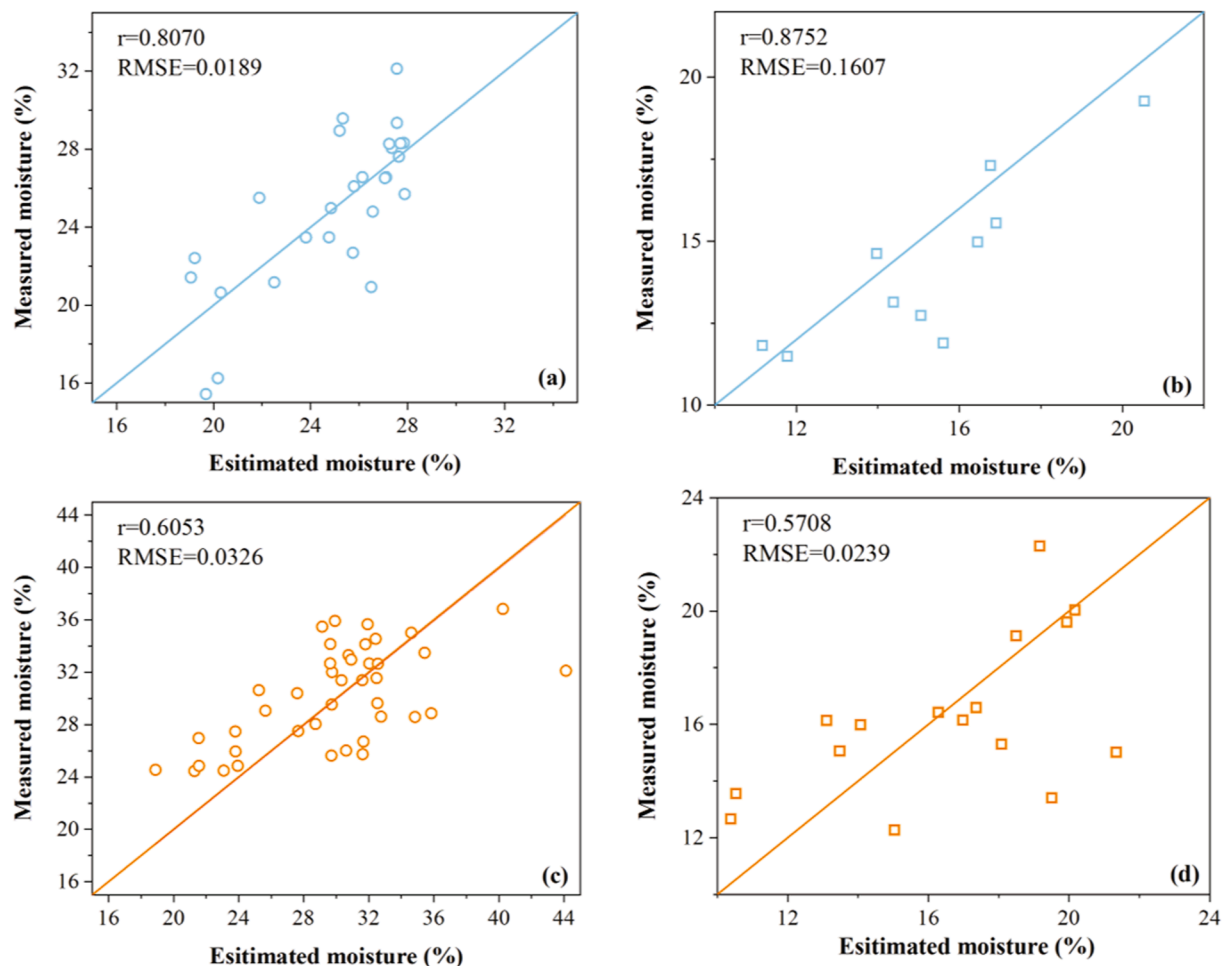


Fig. 3. Comparisons between gravimetrically derived soil water content and that derived by ERT in two soil layers. a: soil depth of 0–40 cm before simulated rainfall; b: soil depth of >40 cm before simulated rainfall; c: soil depth of 0–40 cm after simulated rainfall; d: soil depth of >40 cm after simulated rainfall.

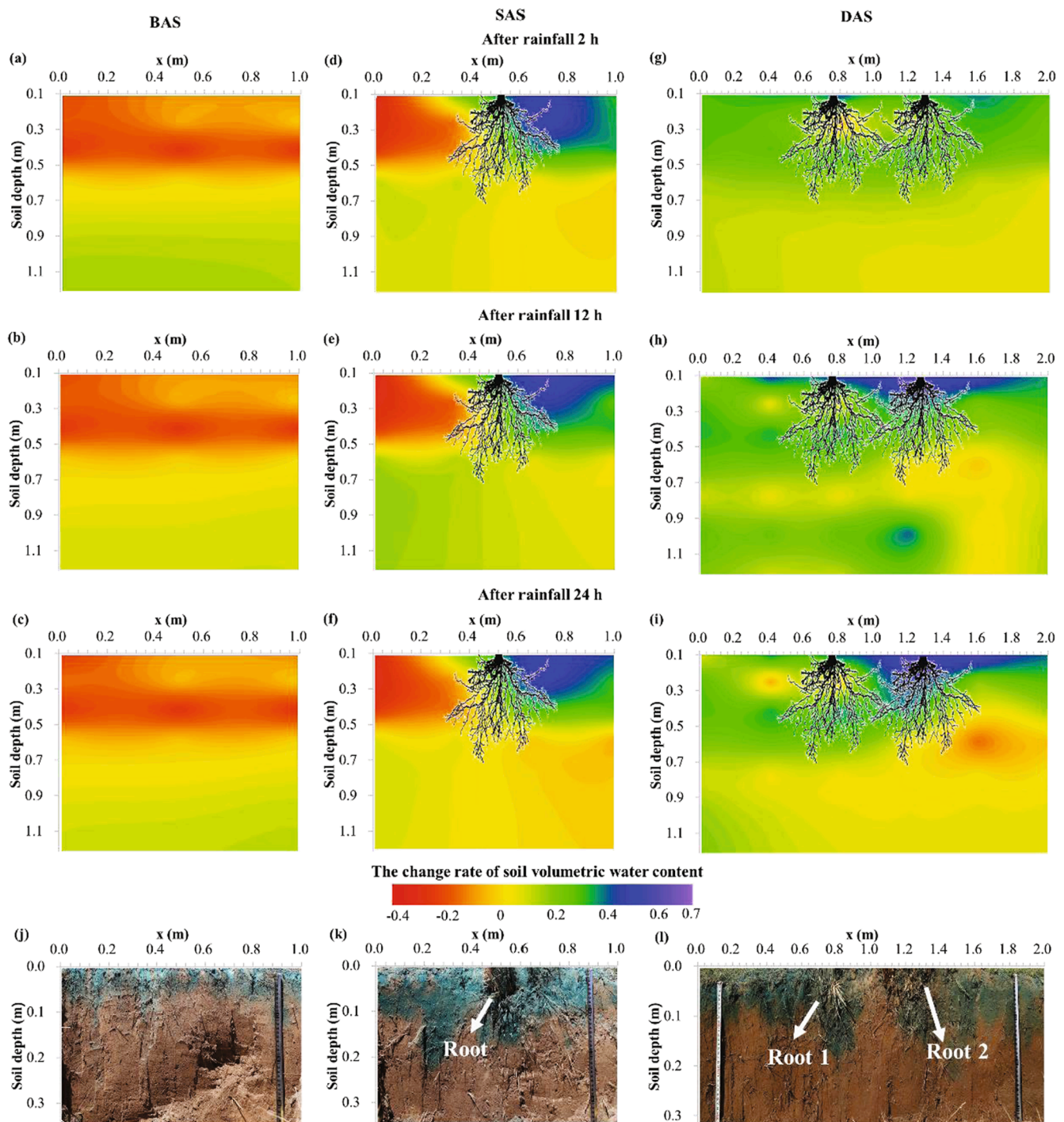


Fig. 4. Vertical transects of the change rate of soil moisture under *A. splendens* with different root densities by ERT (a–c: BAS; d–f: SAS; g–i: DAS) and dyeing profiles (j: BAS; k: SAS; l: DAS) under 20 mm simulated rainfall. Only present a set of data, 20-1BAS, SAS and 20-1DAS, line 3. Black sketches of root indicate root locations of *A. splendens*. BAS is blank *A. splendens*, SAS is single *A. splendens* and DAS is double *A. splendens*.

3.4. Relationship between root traits and soil moisture redistribution

The MID in the staining profile was 38 cm (Fig. 2d). Thus, we focused only on 0–40 cm when analyzing the relationship between the root traits and Δ SVWC after simulated rainfall for 24 h. The relationships determined under SAS and DAS at 0–20 cm included the root traits at depths of 10 and 20 cm, and those at 20–40 cm included the root traits at depths of 30 and 40 cm. The average values of the root traits at 10, 20, 30 and 40 cm were used. The Δ SVWC at 10, 20, 30 and 40 cm derived from ERT were obtained by the linear interpolation method. At a soil depth of

0–20 cm, the relationships between the four root traits and Δ SVWC were all negative and significantly correlated ($P < 0.05$) (Fig. 8a–d), whereas the correlation between RLD and Δ SVWC was extremely significant ($P < 0.001$), and had the highest r of 0.6642 (Fig. 8d). This result indicates that the negative correlation between RLD and Δ SVWC was stronger than those between the other three root traits and Δ SVWC. At a soil depth of 20–40 cm, the RMD and RLD were positively and significantly correlated with Δ SVWC, whereas SRL and SSA were negatively and significantly correlated with Δ SVWC ($P < 0.05$) (Fig. 8e–h). The highest r was 0.5930, between RMD and Δ SVWC (Fig. 8e). This result indicates

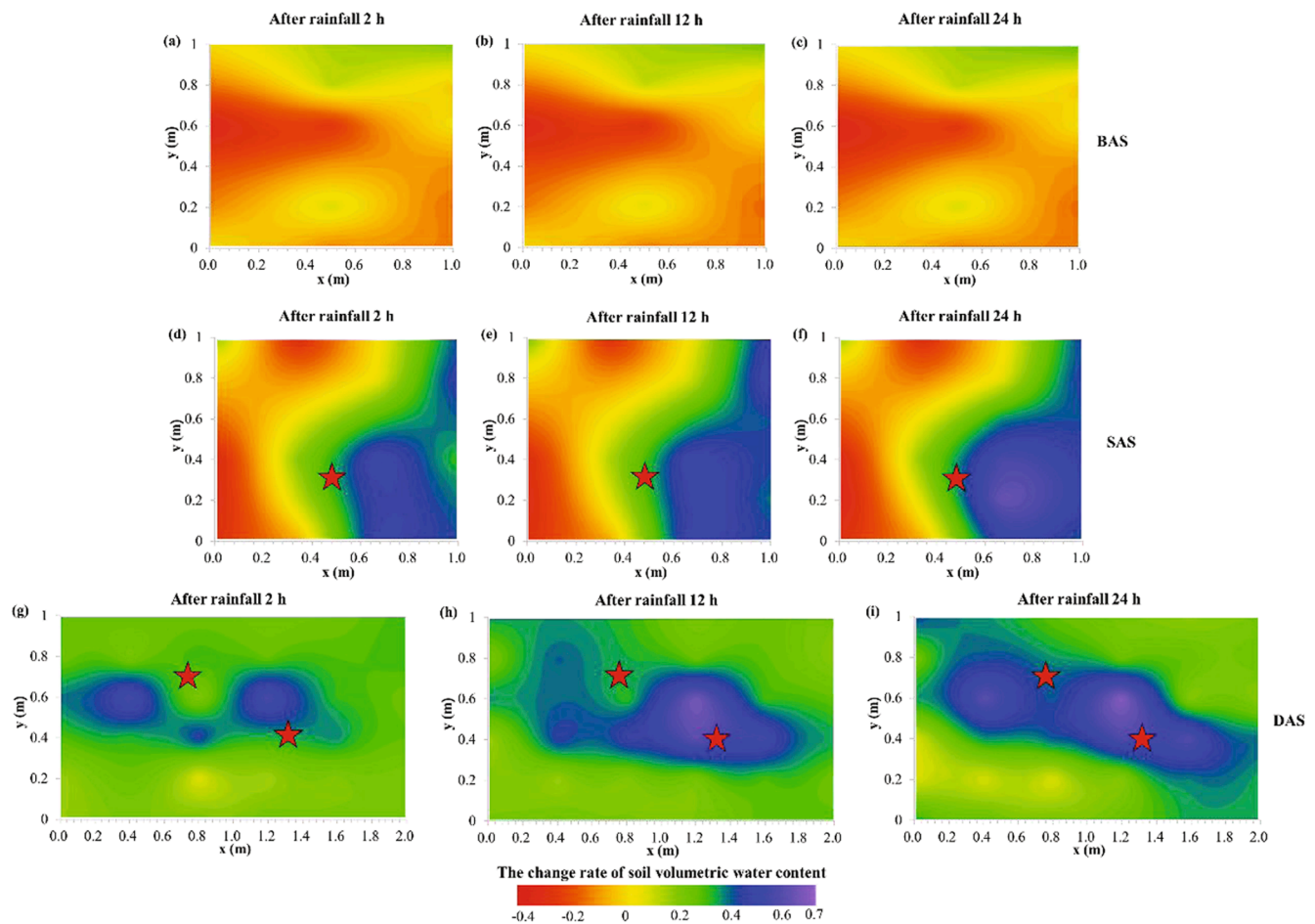


Fig. 5. Temporal dynamics of the change rate of soil moisture at a soil depth of 7.5 cm under 20 mm simulated rainfall (a-c: BAS; d-f: SAS; g-i: DAS). Only present a set of data, 20-1BAS, SAS and 20-1DAS. Red pentagons are root locations of *A. splendens*. BAS is blank *A. splendens*, SAS is single *A. splendens* and DAS is double *A. splendens*.

that the negative correlation between RMD and Δ SVWC was stronger than that between the other three root traits and Δ SVWC.

4. Discussion

The findings of this study highlight that infiltration is closely linked with root distribution. The spatial and temporal variability in the soil moisture data derived from ERT and the dye experiments demonstrated the impact of roots on soil infiltration. For example, the strong increases in soil moisture obviously corresponded to the subsurface root position in the dyeing profiles (Fig. 4f, i, k, l; Fig. S6f, h, j, k; Fig. S9f, i, k, l; Fig. S12f, i, k, l). The root system is the dynamic interface between soil and plants, and its structure significantly affects ecosystem functioning (Helliwell et al., 2017). This study found that the linkages between root traits and soil water change were different at different root densities of *A. splendens* and different soil depths. A subsurface funneling pattern refers to the nonuniform routing of precipitation along roots as a preferential flow path (Schwartzel et al., 2012; Johnson and Lehmann, 2006; Levia and Frost, 2003). Previous studies have suggested that the macropores that form along roots facilitate deeper water flow and faster penetration (Guo et al., 2019; Li et al., 2009; Johnson and Lehmann, 2006). However, our study showed that soil moisture accumulated in the shallow layer (0–20 cm) under SAS (Figs. 4, S6, S9 and S12; Table 3). The relationships between root traits and soil water dynamics in this study can explain this phenomenon (Fig. 8); it may have occurred because the lower RLD of SAS in the shallow layer compared with that of DAS resulted in less soil pores and decreased the vertical soil infiltration

rate (Leung et al., 2015; Archer et al., 2002). *A. splendens*, which does not have main roots in its root system, is different from trees (Jiang, 2016), and the RMD, SRL and SSA of *A. splendens* at 0–20 cm were less than that of 20–40 cm (Fig. 1a, b, d). These root traits also caused more soil moisture to accumulate in the shallow layer (0–20 cm). In addition, the soil type in this study area is a chestnut soil, which easily forms calcium deposits in the soil (Nosalewicz and Lipiec, 2014; Dawson, 1993) that prevent water infiltration to the deep soil. A funneling pattern was observed in DAS (Table 3), which was consistent with the reported conclusions that stemflow along roots can result in a greater increase in soil moisture at deeper depths (Guo et al., 2020; Fan et al., 2015). Numerous studies have shown that the infiltration rate is positively related to the root dry weight density and root biomass (Huang et al., 2019, 2017; Liu et al., 2019). This relationship is mainly attributable to roots forming root channels and large pores in the soil; root channels are beneficial for water transport and substrate flow (Guo et al., 2020; Hu et al., 2020, 2019; Van Schaik, 2009). The DAS root system, with high RMD, might form a larger funnel pattern than the SAS root system, thus overcoming the hindering effect of calcium deposits and allowing macropores to form along the roots to facilitate deeper water flow and faster water penetration (Guo et al., 2019; Li et al., 2009; Johnson and Lehmann, 2006). Several studies have demonstrated that the fine root distribution in the soil profile is positively correlated with the variation in soil water (Zhou and Shangguan, 2007; Powers et al., 2005); these findings are not consistent with the negative relationships of the SRL and SSA with Δ SVWC in this study (Fig. 8b, c, f, g). These negative correlations indicate that the long and thin roots under these

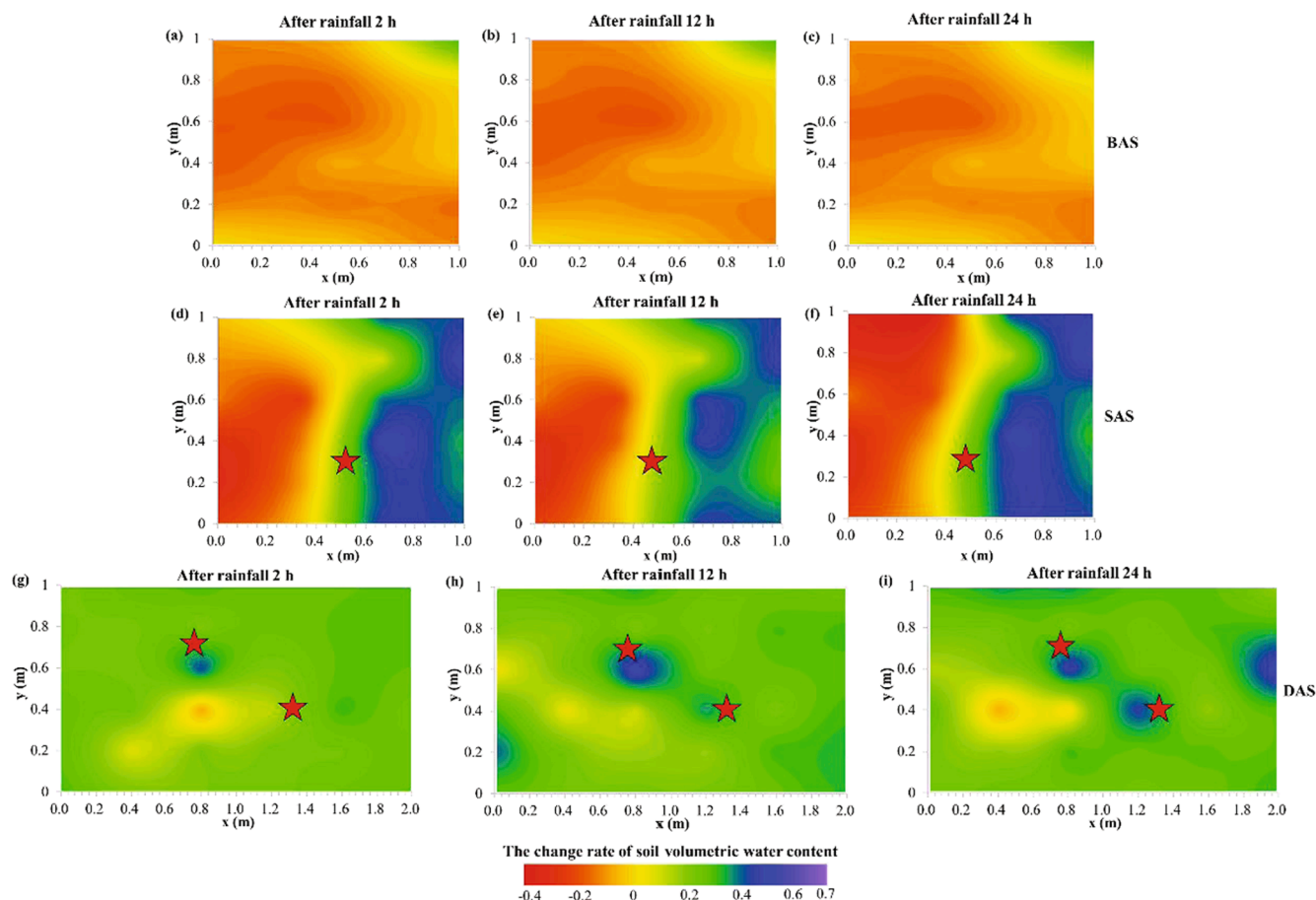


Fig. 6. Temporal dynamics of the change rate of soil moisture at a soil depth of 22.5 cm simulated rainfall (a-c: BAS; d-f: SAS; g-i: DAS) under 20 mm simulated rainfall. Only present a set of data, 20-1BAS, SAS and 20-1DAS. Red pentagons are root locations of *A. splendens*. BAS is blank *A. splendens*, SAS is single *A. splendens* and DAS is double *A. splendens*.

Table 3

The mean values and standard deviations of the soil moisture change rate with time and rainfall amount under different root densities and soil depths.

Soil depth (cm)		BAS		SAS		DAS	
		Mean value	Standard deviation	Mean value	Standard deviation	Mean value	Standard deviation
2 h	7.5	21.83%	0.1990	22.30%	0.1584	14.78%	0.1817
	22.5	15.62%	0.2928	9.50%	0.2680	14.91%	0.1334
	38.3	7.80%	0.3592	22.90%	0.1086	24.83%	0.0316
12 h	7.5	26.88%	0.1682	28.52%	0.1234	22.88%	0.1241
	22.5	23.74%	0.1602	24.24%	0.1023	23.10%	0.0557
	38.3	19.26%	0.2044	22.76%	0.0838	26.30%	0.0194
24 h	7.5	23.16%	0.1982	23.30%	0.1576	22.82%	0.1336
	22.5	15.87%	0.2944	12.20%	0.2868	21.72%	0.0918
	38.3	6.32%	0.3543	18.22%	0.1031	26.07%	0.0244
5 mm	7.5	19.85%	0.0529	14.64%	0.2450	1.19%	0.1843
	22.5	8.55%	0.1235	13.21%	0.4430	30.97%	0.1552
	38.3	0.20%	0.1301	24.48%	0.5395	8.80%	0.0318
10 mm	7.5	34.11%	0.0356	34.61%	0.0198	12.67%	0.0792
	22.5	31.37%	0.0301	28.81%	0.0424	18.66%	0.0453
	38.3	31.57%	0.0434	26.54%	0.1197	24.76%	0.0175
20 mm	7.5	4.84%	0.1605	16.29%	0.0901	35.75%	0.0689
	22.5	5.19%	0.1901	12.11%	0.0899	24.66%	0.0129
	38.3	-9.46%	0.1456	23.41%	0.0127	26.70%	0.0086
40 mm	7.5	33.05%	0.0689	37.51%	0.0419	21.66%	0.0194
	22.5	25.63%	0.0125	28.52%	0.0265	27.58%	0.0357
	38.3	18.57%	0.0764	21.38%	0.0791	28.26%	0.0145

Note: BAS is blank *A. splendens*, SAS is single *A. splendens* and DAS is double *A. splendens*

treatments enhanced the absorption of soil moisture. This is corroborated by the fact that increase in SRL and SSA are derived from root photosynthate, which is used mainly for the elongation of roots or

change in root diameter (Saidi et al. 2010), and demonstrate the ability of these roots to acquire soil resources (de Vries et al., 2016; Poorter and Ryser, 2015). Root biomass is used to grow fine roots, resulting in

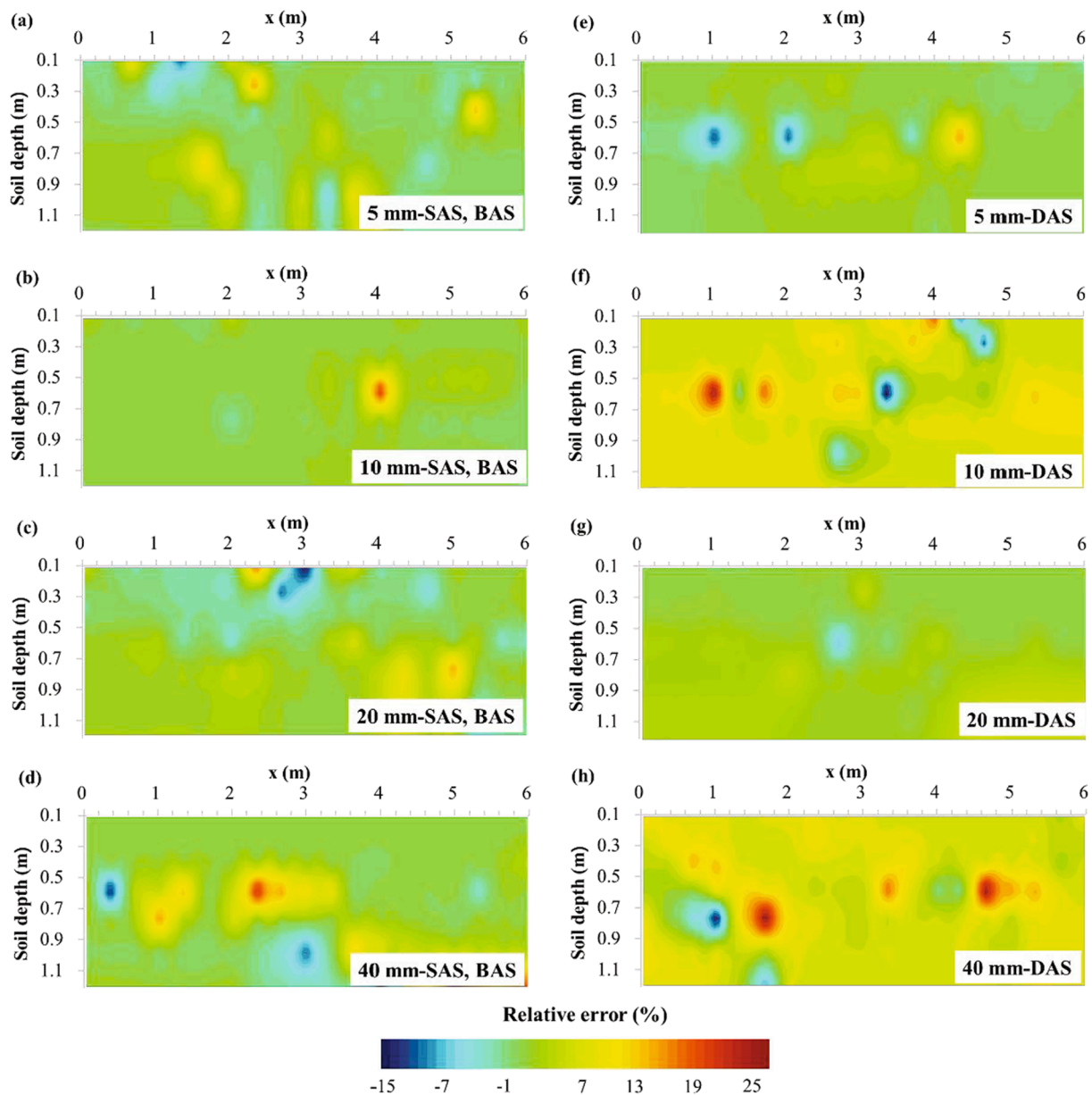


Fig. 7. Average relative error of the change rate of soil moisture between two reduplicated experiments with different simulated rainfall amount for 24 h. a-d: transects of BAS and SAS; e-h: transects of DAS. BAS is blank *A. splendens*, SAS is single *A. splendens* and DAS is double *A. splendens*.

increasing SRL (Hecht et al., 2016) which reduces the contribution of RMD to water infiltration. This supports the negative correlation between the SRL and SSA and the increase in soil water in this study. Previous studies have revealed that increasing the sowing density enhanced the fine root distribution (Hecht et al., 2019), and a higher SRL indicates a higher proportion of fine root length (Li et al., 2017; Ryser, 2006). The results of this study indicate that there were more fine roots in DAS than in SAS, which is consistent with the determined root characteristics of SAS and DAS (Fig. 1). Therefore, the increasing effect of the RMD on soil water is stronger than the decreasing effect of the SRL and SSA on soil water in the deep soil layer; thus, the soil water in deep soil increased under DAS.

This study also found that extreme (40 mm) rainfall events further facilitated the accumulation of water in the shallow soil in SAS and the infiltration of water into deep soil in DAS. The mean Δ SVWC at 38.3 cm in DAS was higher than those in BAS and SAS, whereas the increase rate at 7.5 cm in SAS was higher than those in BAS and DAS under 40 mm simulated rainfall (Table 3). This outcome is consistent with the fact that

the sensitivity of soil moisture to precipitation is different between shallow and deep soil layers (Liu et al., 2017; Chen et al., 2015). The local soil water profiles determined by the precipitation infiltration depth were influenced by climate and soil and had a strong influence on rooting depth (Fan et al., 2017). The deepening of the soil water availability profile as a result of increased precipitation variability may promote deeper-root growth by plants (Sala et al., 2015) and different plants exhibit different rooting strategies in response to water stress (Göransson et al., 2016). In terms of SAS, the shallow soil moisture condition (0–20 cm) favors root system growth under extreme rainfall condition. This result can be attributed to the fact that fine roots may promote resource acquisition by increasing the SRL when precipitation is abundant (Weemstra et al., 2016; Verburg et al., 2013). It is also corroborated by the fact that grasses respond quickly to rainfall, obtain rainfall rapidly from surface soil layers, and thereby gain an advantage when competing with shrubs for water (Archer et al., 2012). Soil moisture affects vegetation growth and can even change weather processes and the local climate by directly participating in soil hydrological

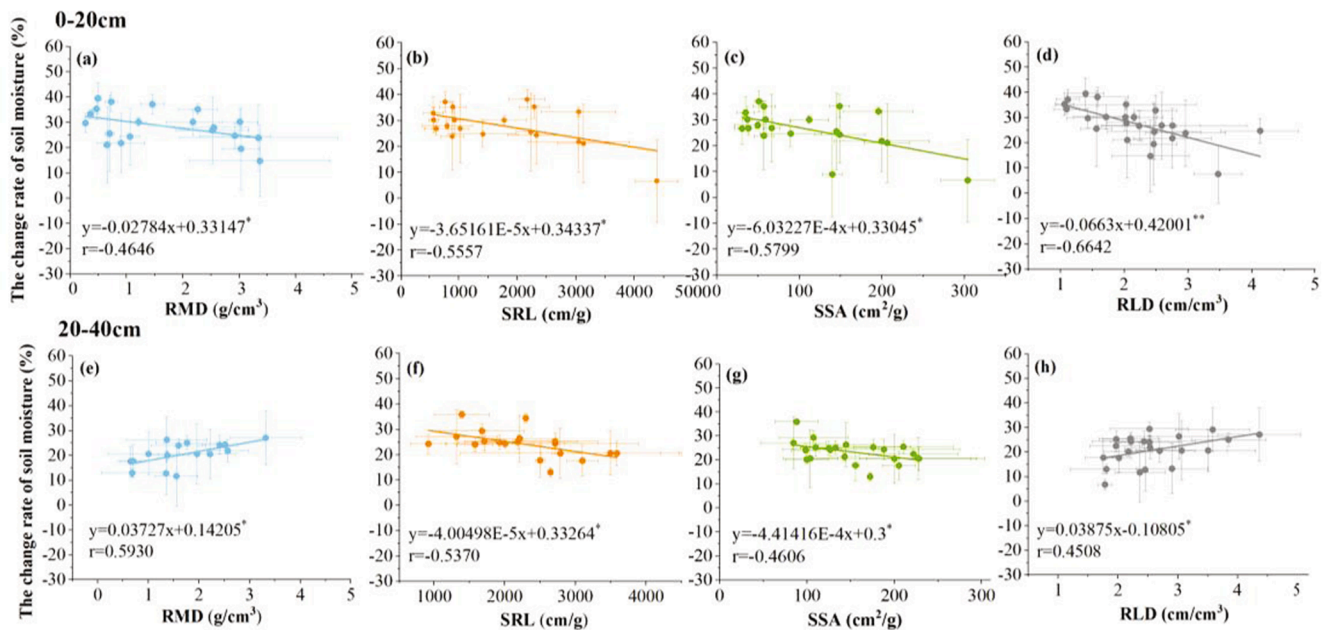


Fig. 8. Relationships between root traits and the change rate of soil moisture (a-d: at soil depths of 0-20 cm; e-h: at soil depths of 20-40 cm). * represents $P < 0.05$; ** represents $P < 0.001$; error bars represent the half standard deviation.

processes ((Legates et al., 2011; Vereecken et al., 2013, 2015; Western et al., 2003)). Therefore, the various linkages between the root traits and Δ SVWC of different plants in arid and semiarid areas might have an impact on future climate change. Regarding the research method in this study, the certainty of the ERT data interpretation improved to some extent by dyeing experiments. However, there are still shortcomings in the determination of instrument accuracy and the construction of petrophysical relationships that need to be addressed in the future.

5. Conclusions

The visualization of ERT and the dyeing profiles in this work showed that the roots of *A. splendens* greatly affected infiltration processes. We quantified the spatiotemporal Δ SVWC in the rhizosphere under different root densities and rainfall patterns and explored the linkages of root traits with Δ SVWC. The soil water vertically and laterally funneled along roots. SAS, which had lower RLD, mainly allowed the soil water to accumulate in the shallow layer (0–20 cm), while DAS, which had higher RMD mainly allowed the soil water to infiltrate rapidly into the deep soil (>20 cm), especially in extreme (40 mm) rainfall events. The increases in soil moisture in deep soil under DAS were higher than those under BAS and SAS after rainfall for 2 and 24 h. The SRL, SSA, RMD and RLD were all negatively and significantly correlated with the Δ SVWC in the shallow soil layer. In deep soil layer, the RMD and RLD were positively and significantly correlated with Δ SVWC, whereas the SRL and SSA showed the opposite trend. These findings provide direct evidence of the linkages between *A. splendens* root traits and Δ SVWC. In the future, the observations of the evapotranspiration and hydraulic conductivity of plants can be supplemented to quantify the relationship between plant water uptake and soil moisture and thereby to improve the study of the SPAC models.

No conflict of interest exists in the submission of this manuscript, and manuscript is approved by all authors for publication. I would like to declare on behalf of my co-authors that the work described was original research that has not been published previously, and not under consideration for publication elsewhere, in whole or in part. All the authors listed have approved the manuscript that is enclosed.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.geoderma.2020.114908>.

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